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3、令 $u=\sum_{n=0}^{\infty}X(x)\cos\left(\frac{n\pi}{b}y\right)$

$\Rightarrow \quad X(x)=C_n\sinh\left(\frac{n\pi}{a}x\right)+D_n\cosh\left(\frac{n\pi}{a}x\right)$

一般解：

$$u=\sum_{n=0}^{\infty}\left[C_n\sinh\left(\frac{n\pi}{a}x\right)+D_n\cosh\left(\frac{n\pi}{a}x\right)\right]\cos\left(\frac{n\pi}{b}y\right)$$

代入初值，

$$\begin{cases} u_0=\sum_{n=0}^{\infty}D_n\cos\frac{n\pi}{b}y \Rightarrow \quad D_n=\frac{2}{b}\int_0^b u_0\cos\frac{n\pi}{b}y\,dy=0\ (n\neq 0),\ D_0=u_0 \\ u_0\left[\left(\frac{y}{b}\right)^2-2\frac{y}{b}\right]=\sum_{n=0}^{\infty}C_n\cos\left(\frac{n\pi}{b}y\right)\Rightarrow C_n=2\int_0^1\left[t^2-2t\right]\cos n\pi t\,dt=\frac{8}{n\pi} \end{cases}$$

综上，$u=\sum_{n=1}^{\infty}\frac{8}{n\pi}\sinh\left(\frac{n\pi}{a}x\right)\cos\frac{n\pi}{b}y+u_0$

4、其满足，

$$\begin{cases} \frac{du}{dt}-D\frac{d^2u}{dx^2}=\beta u \\ u|_{x=0}=u|_{x=L}=0 \end{cases}$$

令 $u=\sum_{n=1}^{\infty}T(t)\sin\frac{n\pi}{L}x$，$\beta u=\sum_{n=1}^{\infty}g_n(t)\sin\frac{n\pi}{L}x$

代入，$T'(t)+D\frac{n^2\pi^2}{L^2}T(t)=g_n(t)=\beta T(t)$

$\Rightarrow \quad T'(t)+\left(\frac{n^2\pi^2}{L^2}D-\beta\right)T(t)=0$

$\Rightarrow \quad T(t)=e^{-\left[\frac{n^2\pi^2}{L^2}D-\beta\right)t}$

即：$u=e^{-\left(\frac{n^2\pi^2}{L^2}D-\beta\right)t}\sin\frac{n\pi}{L}x$

当 $\beta-\frac{n^2\pi^2}{L^2}D>0$，即 $L>\sqrt{\frac{n^2\pi^2D}{\beta}}_{min}=\sqrt{\frac{D}{\beta}}\pi$ 时，火药炸

综上，$L=\sqrt{\frac{D}{\beta}}\pi$

为主方程时，

$$\begin{cases} \dfrac{du}{dt} - D\left[\dfrac{d^2u}{dx^2} + \dfrac{d^2u}{dy^2}\right] = \beta u \\ u|_{x=0} = u|_{y=0} = u|_{x=l} = u|_{y=l} = 0 \end{cases}$$

令 $u = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} T(t)\sin\frac{n\pi}{l}x\sin\frac{m\pi}{l}y$

$$\Rightarrow T(t)' + 2\left(\frac{n\pi}{l}\right)^2 D\,T(t) = \beta T(t)$$

同理，$T(t) = e^{\left(\beta - \frac{2n^2\pi^2}{l^2}D\right)t}$

当 $\beta > \frac{2n^2\pi^2}{l^2}D$，即 $l > \left(\sqrt{\frac{2D}{\beta}}n\pi\right)_{min} = \sqrt{\frac{2D}{\beta}}\pi$ 时，爆炸

$\therefore l_{max} = \sqrt{\frac{2D}{\beta}}\pi$

5、

令 $u = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} C_{mn}\sin\frac{m\pi}{a}x\sin\frac{n\pi}{b}y$，$x^2y = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} d_{mn}\sin\frac{m\pi}{a}x\sin\frac{n\pi}{b}y$

$$\Rightarrow d_{mn} = \frac{4}{ab}\int_0^a\int_0^b x^2y\sin\frac{m\pi}{a}x\sin\frac{n\pi}{b}y\,dx\,dy = \frac{4(n\pi+1)}{(n\pi)^3}a^2$$

又由题意，

$$C_{mn} = -\frac{d_{mn}}{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} = -\frac{4(n\pi+1)}{(n\pi)^5(a+b)}a^3b$$

$$\therefore u = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} -4\frac{n\pi+1}{(n\pi)^5\cdot(a+b)}a^3b\sin\frac{m\pi}{a}x\sin\frac{n\pi}{b}y$$

6、

$$\begin{cases} \dfrac{d^2u}{dt^2} - a^2\dfrac{d^2u}{dx^2} = 0 \\ u|_{x=0} = 0,\ \dfrac{du}{dx}\Big|_{x=l} = A\sin\omega t \\ u|_{t=0} = x,\ \dfrac{du}{dt}\Big|_{t=0} = 0 \end{cases}$$

令 $u = v(x,t) + w(x,t)$，其中：

$$v = f(x)\sin\omega t$$

$$\Rightarrow \begin{cases} f''(x) + \left(\frac{\omega}{a}\right)^2 f(x) = 0 \\ f(0) = 0,\ f'(l) = A \end{cases}$$

得：

$$f(x) = \frac{Aa}{\omega}\cdot\frac{1}{\cos(\omega l/a)}\sin\frac{\omega}{a}x$$

$w(x,t)$ 有：

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} - a^2\frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0} = 0 \\ \frac{\partial w}{\partial x}\Big|_{x=l} = 0 \end{cases}$$

$$\Rightarrow w(x,t) = \sum_{n=0}^{\infty}\left[C_n\sin\frac{2n+1}{2l}\pi a t + D_n\cos\frac{2n+1}{2l}\pi a t\right]\sin\frac{2n+1}{2l}\pi x$$

由初值得，

$$C_n = (-1)^n\frac{16A\omega a}{(2n+1)\pi}\cdot\frac{1}{\omega^2 - [(2n+1)a\pi/2l]^2}$$

$$D_n = 0$$

综上，

$$u(x,t) = v(x,t) + w(x,t)$$

$$= \frac{Aa}{\omega}\cdot\frac{1}{\cos(\omega l/a)}\sin\frac{\omega}{a}x\sin\omega t + (-1)^n\frac{16A\omega a}{(2n+1)\pi}\cdot\frac{1}{\omega^2 - [(2n+1)a\pi/2l]^2}$$